

The St. Petersburg

I am going to flip a fair coin until it comes up heads. If the first time it comes up heads is on the 1st toss, I will give you \$2. If the first time it comes up heads is on the second toss, I will give you \$4. If the first time it comes up heads is on the 3rd toss, I will give you \$8. And in general, if the first time the coin comes up heads is on the n th toss, I will give you $\$2^n$.

Would you pay \$2 to take this bet? How about \$4?

Suppose now I raise the price to \$10,000. Should you be willing to pay that amount to play the game once?

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